

ZIXIAN LIU

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EDUCATION

Purdue University

B.S. in Computer Science; Minor in Mathematics (GPA: 3.94/4.0)

West Lafayette, IN

Jul. 2023 – May 2027 (expected)

- **Selected coursework:** Machine Learning, Algorithms, Operating Systems, Numerical Methods, Linear Algebra, Statistics.

MANUSCRIPTS UNDER REVIEW

- [1] First-author manuscript on real-world robot learning and manipulation, under review at **IEEE ICRA 2027**. (*Title omitted during double-anonymous review.*)

RESEARCH EXPERIENCE

Machine Learning and Robotics Research, Prof. Yung-Hsiang Lu

Jan. 2026 – present

Purdue University

Multimodal Robot Learning with Action-Space Expansion and Cross-String Adaptation

Aug. 2026 – present

- Developing a policy architecture that integrates multimodal sensory observations, scores from a supervised human-preference model, and musical score requirements to generate bowing commands for a UR5e-based robotic cellist.
- Investigating multimodal reward design and evaluation using audio, force/torque, vision, and robot state, with independent listening studies to assess performance.
- Extending the bowing action space from speed and depth to include bow angle and contact position.
- Conducting experiments to assess how learned bowing policies generalize across cello strings and adapt to string-specific contact conditions.

Residual RL for Contact-Rich Bowing [1]

Jan. 2026 – Sep. 2026

- Developed a four-stage residual SAC framework for a UR5e robotic cellist under contact drift, using a fixed reward classifier trained on human ratings.
- Designed held-out classifier evaluation and a blinded listening study. Policies achieved a 15 % mean held-out classifier-score gain over the planner across five pieces.
- Built the real-robot training and safety stack; wrote the first-author manuscript and an audit reproducing reported results from recordings and logs.

Team lead since Summer 2026; coordinate five undergraduates across two research tracks in experiment planning and results analysis; attend the weekly graduate research meeting.

SELECTED PROJECTS

Xinu OS

Aug. 2026 – present

- Implemented semaphore-based synchronization and an independent IPC extension with bounded message buffers and blocking send/receive. Deployed and tested Xinu on ARM-based BeagleBone Black hardware.
- Implemented a kernel-level semaphore checker to detect inconsistencies in semaphore counts, waiting queues, and process states.

Unix Shell

Jan. 2026 – May 2026

- Built a bash-style shell in C/C++ with a lex/yacc parser, supporting pipes, redirection, background jobs, subshells, signal handling, variable and wildcard expansion, and command history.

Systematic Trend-Timing Strategy

Feb. 2026 – Sep. 2026

- Built a vectorized backtester using 100 years of daily data, modeling next-open execution, transaction costs, and taxes.
- Evaluated 233 strategy variants using pre-registered walk-forward validation, block bootstrap, and multiple-testing correction; identified selection overfitting and reduced crisis drawdowns without excess returns.

TECHNICAL SKILLS

Programming

Python, Java, C/C++, R, MATLAB, SQL

Robot learning / ML

PyTorch, SAC, PPO, residual reinforcement learning, learned reward models

Robotics

UR5e / ur_rtde, force/torque sensing, contact calibration, safety supervisors

Research tools

Linux, Git, L^AT_EX